

HuggingFace Spaces Deployment Guide

Complete guide for deploying CodeForge AI to HuggingFace Spaces.

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Prerequisites

Required

- **HuggingFace Account:** [Sign up here](https://huggingface.co/join) (<https://huggingface.co/join>)
- **HuggingFace API Token:** [Get token here](https://huggingface.co/settings/tokens) (<https://huggingface.co/settings/tokens>)
- **Python 3.8+:** For running deployment scripts

Optional (for full functionality)

- **Venice AI API Key:** For AI code generation ([Venice AI](https://venice.ai) (<https://venice.ai>))
 - **GitHub Personal Access Token:** For GitHub integration ([Create token](https://github.com/settings/tokens) (<https://github.com/settings/tokens>))
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Quick Start

One-Command Deployment

```
# Install dependencies
pip install huggingface_hub

# Set your HuggingFace token
export HF_TOKEN="your-huggingface-token"

# Deploy (creates a new Space)
python deploy_to_huggingface.py --space-name my-codeforge-ai

# Or deploy lightweight Gradio version
python deploy_to_huggingface.py --space-name my-codeforge-ai --gradio-only
```

Your Space will be available at: https://huggingface.co/spaces/YOUR_USERNAME/my-codeforge-ai

Deployment Options

Option 1: Full Deployment (Next.js + FastAPI)

Best for: Full-featured experience with rich UI

```
python deploy_to_huggingface.py --space-name codeforge-full
```

Includes:

- Next.js React frontend with Monaco editor
- FastAPI backend with all features
- WebSocket support for real-time updates
- Full project management capabilities

Resource Usage:

- Higher memory usage (~4-8GB)
- Longer build time (~5-10 minutes)

Option 2: Gradio-Only Deployment

Best for: Free tier, quick deployment, simple interface

```
python deploy_to_huggingface.py --space-name codeforge-lite --gradio-only
```

Includes:

- Lightweight Gradio interface
- Core AI chat and code generation
- Project and agent management
- Chat parsing functionality

Resource Usage:

- Lower memory usage (~2-4GB)
- Faster build time (~2-5 minutes)

Step-by-Step Deployment

Step 1: Clone the Repository

```
git clone https://github.com/your-username/codeforge-ai.git
cd codeforge-ai
```

Step 2: Install Deployment Dependencies

```
pip install huggingface_hub
```

Step 3: Authenticate with HuggingFace

```
# Option A: Set environment variable
export HF_TOKEN="hf_your_token_here"

# Option B: Login interactively
huggingface-cli login
```

Step 4: Run Deployment Script

```
# Basic deployment
python deploy_to_huggingface.py --space-name my-codeforge

# With all options
python deploy_to_huggingface.py \
  --space-name my-codeforge \
  --private \
  --sync-secrets \
  --hardware cpu-basic
```

Step 5: Configure Secrets

After deployment, go to your Space settings and add:

1. Navigate to: `https://huggingface.co/spaces/YOUR_USERNAME/my-codeforge/settings`
2. Scroll to “Repository secrets”
3. Add:
 - `VENICE_API_KEY` : Your Venice AI key
 - `GITHUB_PERSONAL_ACCESS_TOKEN` : Your GitHub token
 - `SECRET_KEY` : A random secure string

Step 6: Verify Deployment

1. Wait for build to complete (check “Building” status)
 2. Once “Running”, click your Space URL
 3. Test the interface
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Configuration

Deployment Script Options

Option	Description	Default
<code>--space-name</code>	Name for your HF Space	<code>codeforge-ai</code>
<code>--username</code>	HuggingFace username	Auto-detected
<code>--token</code>	HF API token	<code>HF_TOKEN</code> env var
<code>--private</code>	Make Space private	<code>false</code>
<code>--gradio-only</code>	Deploy Gradio version	<code>false</code>
<code>--update</code>	Update existing Space	<code>false</code>
<code>--sync-secrets</code>	Sync .env to HF secrets	<code>false</code>
<code>--hardware</code>	Hardware tier	<code>cpu-basic</code>
<code>--no-monitor</code>	Skip build monitoring	<code>false</code>

Hardware Tiers

Tier	Memory	Cost	Best For
<code>cpu-basic</code>	16GB	Free	Gradio version
<code>cpu-upgrade</code>	32GB	~\$0.06/hr	Full version
<code>t4-small</code>	GPU	~\$0.60/hr	Local LLM
<code>t4-medium</code>	GPU	~\$0.90/hr	Faster LLM

Environment Variables

Secrets (Sensitive - Set in HF Secrets)

```
VENICE_API_KEY=your-venice-api-key
GITHUB_PERSONAL_ACCESS_TOKEN=ghp_your_github_token
SECRET_KEY=your-random-secret-key-minimum-32-chars
```

Variables (Non-sensitive - Set in HF Variables)

```
DATABASE_URL=sqlite:///app/data/codeforge.db
USE_SQLITE=true
HOST=0.0.0.0
PORT=7860
DEBUG=false
GITHUB_USERNAME=your-github-username
DEFAULT_AI_PROVIDER=venice
LOG_LEVEL=INFO
```

How to Set Variables

1. Go to Space Settings
2. Find “Repository secrets” section
3. Click “New secret” for sensitive values
4. Find “Variables” section for non-sensitive values

Database Options

Option 1: SQLite (Recommended for Free Tier)

SQLite is automatically configured for HF Spaces:

```
# Automatically used when USE_SQLITE=true
DATABASE_URL=sqlite:///app/data/codeforge.db
```

Pros:

- No external dependencies
- Works on free tier
- Persistent within Space

Cons:

- Limited concurrent access
- Data lost if Space is deleted

Option 2: External PostgreSQL

For production use, connect to an external PostgreSQL:

```
DATABASE_URL=postgresql://user:password@host:5432/database
USE_SQLITE=false
```

Recommended Providers:

- [Supabase](https://supabase.com) (https://supabase.com) - Free tier available
- [Railway](https://railway.app) (https://railway.app) - Easy setup
- [Neon](https://neon.tech) (https://neon.tech) - Serverless PostgreSQL

Migrating from PostgreSQL to SQLite

```
python backend/app/database/sqlite_adapter.py \
  --migrate "postgresql://user:pass@host:5432/db" \
  --output /app/data/codeforge.db
```

Troubleshooting

Build Failures

Problem: Space shows “Build Error”

Solutions:

1. Check build logs in Space settings
2. Verify Dockerfile syntax
3. Ensure all dependencies are in requirements.txt
4. Check for missing files

```
# Test locally first
docker build -t codeforge-test .
docker run -p 7860:7860 codeforge-test
```

Runtime Errors

Problem: Space crashes after starting

Solutions:

1. Check runtime logs
2. Verify environment variables are set
3. Ensure database directory is writable
4. Check memory usage (free tier: 16GB max)

API Connection Issues

Problem: AI chat returns errors

Solutions:

1. Verify `VENICE_API_KEY` is set correctly
2. Check API key is valid (test with curl)
3. Ensure no rate limiting

```
# Test Venice API
curl -X POST https://api.venice.ai/api/v1/chat/completions \
  -H "Authorization: Bearer YOUR_API_KEY" \
  -H "Content-Type: application/json" \
  -d '{"model": "dolphin-2.9.2-qwen2-72b", "messages": [{"role": "user", "content": "Hello"}]}'
```

Database Issues

Problem: “Database locked” or connection errors

Solutions:

1. Restart the Space
 2. Check SQLite file permissions
 3. For high traffic, consider external PostgreSQL
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 Updating Your Space

Quick Update

```
# Update existing Space
python deploy_to_huggingface.py --space-name my-codeforge --update
```

Using Update Script

```
# Make executable
chmod +x update_space.sh

# Run update
./update_space.sh --space-name my-codeforge

# Update Gradio version
./update_space.sh --space-name my-codeforge --gradio

# Update with secrets
./update_space.sh --space-name my-codeforge --sync-secrets
```

Manual Update via Git

```
# Clone your Space
git clone https://huggingface.co/spaces/YOUR_USERNAME/my-codeforge
cd my-codeforge

# Make changes
# ... edit files ...

# Push updates
git add .
git commit -m "Update feature X"
git push
```

 Cost Considerations

Free Tier

Feature	Included
CPU	Basic (2 vCPU)
Memory	16GB
Storage	50GB
Bandwidth	Unlimited
Uptime	Sleeps after inactivity

Note: Free tier Spaces sleep after ~15 minutes of inactivity. First request after sleep takes ~30-60 seconds.

Paid Tiers

Tier	Cost	Benefits
CPU Upgrade	~\$0.06/hr	8 vCPU, 32GB RAM, no sleep
T4 Small	~\$0.60/hr	GPU, good for local LLM
T4 Medium	~\$0.90/hr	GPU, better for larger models

Cost Optimization Tips

1. Use Gradio version for free tier
 2. Disable local LLM (uses external Venice AI)
 3. Set shorter inactivity timeout
 4. Use external database for persistence
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Limitations

Free Tier Limitations

Limitation	Impact	Workaround
CPU only	No local GPU LLM	Use Venice AI
16GB memory	Limited concurrent users	Use Gradio version
Sleep after inactivity	Slow first request	Upgrade hardware
No persistent storage guarantee	Data may be lost	External database
Build time limit	Large builds may fail	Optimize Dockerfile

Feature Limitations on HF Spaces

Feature	Status	Notes
AI Chat	✓ Full	Via Venice AI
Code Generation	✓ Full	Via Venice AI
Local LLM	⚠ Limited	Needs GPU tier
Project Management	✓ Full	SQLite storage
GitHub Integration	✓ Full	Set token
Real-time WebSocket	⚠ Limited	May disconnect
File Export	✓ Full	Works normally

FAQ

Q: How do I get a HuggingFace token?

1. Go to [HuggingFace Settings](https://huggingface.co/settings/tokens) (https://huggingface.co/settings/tokens)
2. Click “New token”
3. Select “Write” access
4. Copy the token

Q: Can I use my own domain?

Yes, HuggingFace supports custom domains for paid accounts. Contact HF support.

Q: How do I backup my data?

```
# Export projects via API
curl https://your-space.hf.space/api/projects/export -o backup.zip

# Or use the Export feature in the UI
```

Q: Can I run this locally instead?

Yes! Use docker-compose:

```
docker-compose up --build
# Access at http://localhost:7860
```

Q: Why is my Space slow to start?

Free tier Spaces sleep after inactivity. The first request wakes it up, which takes 30-60 seconds.

Q: How do I delete my Space?

1. Go to Space settings
2. Scroll to “Danger Zone”
3. Click “Delete this Space”



Additional Resources

- [HuggingFace Spaces Documentation](https://huggingface.co/docs/hub/spaces) (https://huggingface.co/docs/hub/spaces)
- [Docker SDK Guide](https://huggingface.co/docs/hub/spaces-sdks-docker) (https://huggingface.co/docs/hub/spaces-sdks-docker)
- [Venice AI Documentation](https://docs.venice.ai) (https://docs.venice.ai)
- [CodeForge AI GitHub](https://github.com/your-username/codeforge-ai) (https://github.com/your-username/codeforge-ai)



Support

If you encounter issues:

1. Check the [Troubleshooting](#) section
2. Search [HuggingFace Forums](https://discuss.huggingface.co) (https://discuss.huggingface.co)
3. Open an issue on [GitHub](https://github.com/your-username/codeforge-ai/issues) (https://github.com/your-username/codeforge-ai/issues)

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